Shinigami Eyes — Project Brief

Oct 19, 2026 · @David

Shinigami Eyes is a private, Watch Dogs-style profiler of David's surroundings: a full-screen iPhone app backed by his Proxmox server. It is built over the summer holidays, after exams on 2–4 Nov 2026, as a separate project from SPENCER.

Vision

Open the app anywhere and see a live profile of where you are: the street, the room, the sky, the airwaves, and yourself. It profiles places, infrastructure and David's own life, never other people.

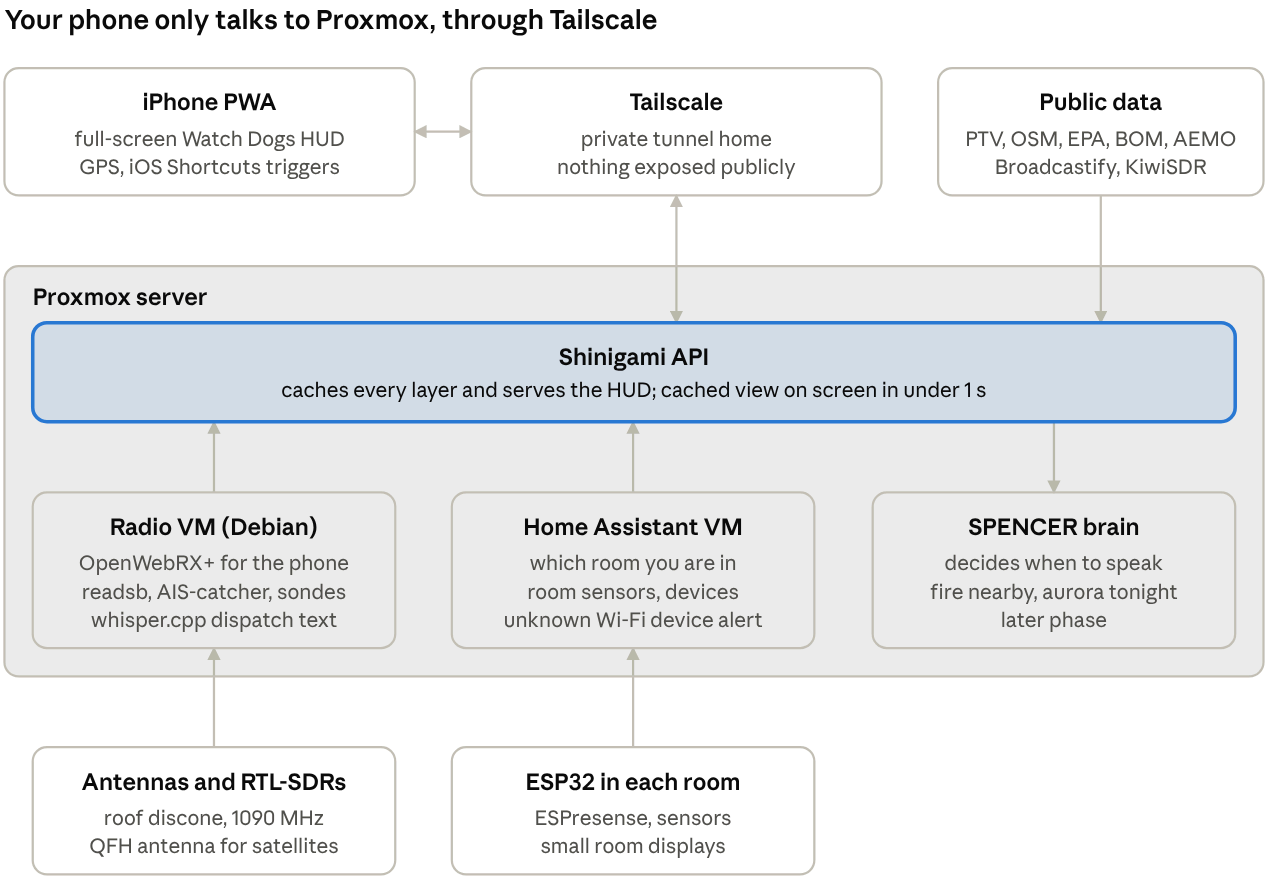
Design rule: like the Shinigami eyes in Death Note, everything shown gets a name and a countdown.

- Tram: arrives in 3:12
- Exam: 14 days
- Street tree: River Red Gum, useful life 40+ years
- E: drive: full in 23 days
- Golden hour: ends in 18 min
- Gym streak: dies at midnight
- L-plate hours: 47 of 120 logged

First screen, the self-profile card: DAVID, 20, Chem Eng at Monash. Exams, gym streak, sleep, last call to Mum, server health, next tram.

Relationship to SPENCER and LifeOS: Shinigami is the eyes, SPENCER is the voice. Shinigami detects things; SPENCER decides whether to interrupt. LifeOS is the existing life dashboard; how Shinigami feeds it, or reads from it, is an open question.

Architecture



architecture · phone, tunnel, server, sources

Everything runs on Proxmox; the phone reaches it only through Tailscale, and public data, radios and room sensors all feed one API that keeps the HUD fast.

Layers

City mode ships first because it needs no hardware; house mode and radio follow once the ESP32s and SDR dongles arrive.

Layer	What it shows	Data source	Phase
PTV live	Nearest stops, real-time departures, disruptions, vehicles approaching	PTV Timetable API (key by request), Victorian GTFS-realtime	1

Layer	What it shows	Data source	Phase
Surveillance	Known CCTV within ~150 m, fixed speed and red-light cameras	OpenStreetMap surveillance tags via Overpass; Victorian road safety camera list	1
Environment	Air quality, UV, pollen and thunderstorm asthma risk, weather, rain radar, fire danger	EPA Victoria, ARPANSA, Melbourne Pollen, BOM, CFA	1
City of Melbourne	Pedestrian counts, parking bay sensors, street trees, development activity	City of Melbourne open data	1
Buildings	History and context of what is in front of you	Victorian Heritage Database, Wikipedia geosearch	1
Hazards	Fires, floods, incidents near you	VicEmergency (no official public feed yet), AusAlert on the phone itself	1
Sky	Planes overhead, ISS and satellite passes	OpenSky / adsb.lol, CelesTrak; later own ADS-B receiver	1, then 3
Bay	Ships in Port Phillip	aisstream.io; later own AIS receiver	1, then 3
Towers	Licensed transmitter sites and carriers near you	ACMA Register of Radiocommunications Licences	1
Grid and water	Victoria's live generation mix and price, rooftop solar share, water storage	AEMO via OpenElectricity, Melbourne Water	1
Industry	Licensed industrial premises and what they may emit	EPA Victoria licence register	2
House	Which room you are in, room conditions, devices on, registered household home, unknown devices on Wi-Fi	ESP32 + ESPresense + sensors, Home Assistant VM	4

Radio

The radio layer lets you hear the city: receive-only, unencrypted signals, all kept private behind Tailscale. The headline feature is CFA dispatch audio transcribed on Proxmox and matched to your GPS, so SPENCER can say a grass fire is 3 km north of you.

Horizon	What you get	How	Notes
Emergency	CFA and FRV dispatch audio, proximity alerts	Broadcastify feeds (CFA District 14, FRV metro, RRDS); own receiver where in the clear; whisper.cpp transcription	Victoria Police is encrypted. Pager data is off-limits. Check RadioReference for your district's mode
Emergency broadcast	Official warnings	ABC Radio Melbourne 774 AM stream; AusAlert cell broadcast on the iPhone	AusAlert targeted full operation by Oct 2026
Aviation	ATC voice; live plane map incl. air ambulance and police helicopters	Airband (Melbourne Tower 120.5 MHz); readsb + tar1090 on 1090 MHz	Feeding adsb.lol or FlightAware can earn perks
Marine	Channel 16 distress, port traffic, ships	VHF marine; AIS-catcher on 162 MHz	
Space	Weather satellite photos of Victoria from your roof	Meteor-M LRPT on 137 MHz, SatDump, DIY QFH antenna	NOAA APT ended in 2025. Later: GK-2A geostationary with a small dish
Balloons	Track and recover BOM weather balloons	radiosonde_auto_rx around 400 MHz, SondeHub	BOM does not recover them
Broadcast	Song titles, DAB+ slideshows, world shortwave	FM RDS, DAB+ (welle.io), KiwiSDR/WebSDR in the browser	No hardware needed for KiwiSDR
Lightning	Strikes near you in summer storms	Blitzortung data; later own detector	

Horizon	What you get	How	Notes
Amateur	APRS positions and weather stations; later transmit	APRS decoding; Foundation licence; join WICEN Victoria	Class licence, no annual fee; one-off certificate and callsign costs

Server side: a Debian VM on Proxmox with the dongles passed through over USB. OpenWebRX+ gives a full radio in the iPhone browser, but stops listening when the tab closes, so the background decoders run as separate always-on services. One dongle per always-on job. whisper.cpp runs on CPU; if the Quadro M4000 is used, pin CUDA 12, never 13.

Personal layers

Every personal layer stays on Proxmox and never goes to a cloud service.

Layer	Trigger	What it shows	Data
Gym	Shortcuts arrival automation at Hammers Gym Nunawading, or an NFC tag on the gym bag	Today's program, last session's numbers per lift, overload target, PRs, rest timer, streak countdown	Liftoff (export or API unknown); Apple Health via Shortcuts; recovery from sleep if an Apple Watch is bought
Uni	On campus, or the calendar	Next class, room and walk time, Moodle deadlines, exam countdowns, study hours	Moodle calendar export, Google Calendar
Engineer's lens	Anywhere	Live grid mix and price, licensed industrial sites nearby, air chemistry, water storage	AEMO via OpenElectricity, EPA Victoria, Melbourne Water

Layer	Trigger	What it shows	Data
Learner driver	Driving sessions	Logbook hours toward 120 (incl. 20 at night), earliest P-test date, road cameras on the route	Own log; check VicRoads for the exact rules
Photography	Light changes, alerts	Golden and blue hour countdowns, aurora australis alerts, ISS passes, cloud cover, map of your own past shots	BOM Space Weather, CelesTrak, BOM, photo GPS data
Home lab	Walk into the server room	Proxmox node and VM health, temperatures, SPENCER uptime, internet latency, Tailscale peers	Proxmox API, SPENCER telemetry
Downtime	When not gaming	Friends online, wishlist sales, patch notes; goes quiet while the GPU is busy	Game platform APIs
Money (optional)	On request only	This week's spending	Only if your bank offers a personal API, and only on Proxmox

Hard rules

Published or broadcast-to-everyone data is in; anything that identifies or tracks a person is out. This is a working reading of the sources, not legal advice.

- No face recognition, number-plate lookups, or logging other people's Wi-Fi or Bluetooth devices.
- People appear only if they opted in: registered household phones at home.
- Receive-only radio, unencrypted signals only. No police decryption, no pager decoding, no phone or cordless interception.

- Nothing rebroadcast or published. The app and every stream stay private behind Tailscale.
- No transmitting until the Foundation licence is held.
- Personal and financial data never leaves Proxmox.
- Never the slow path: cached data on screen in under a second, live data fills in after.

Shopping list

Order now so it lands after exams; prices are rough AUD estimates, check before buying.

Item	Rough cost	For
RTL-SDR Blog V4 kit with dipole, x2	$60–70 each	One for ADS-B around the clock, one for OpenWebRX+
1090 MHz antenna and filtered LNA	$30–50	Plane range
Discone or broadband antenna, roof-mounted	$80–150	Airband, marine, emergency, balloons
QFH antenna parts (DIY)	~$30	Meteor-M satellites
Powered USB hub and short LMR-240 coax	~$40	Keep dongles near the antennas
ESP32 boards with temp/humidity/CO2/motion sensors, per room	~$15 per room	House mode
Small ESP32 display	~$15–25	Room HUD: planes overhead, nearest fire
Later: Airspy or SDRplay	~$300	Wide-band reception
Maybe: Apple Watch	—	Sleep, recovery, heart rate for the gym layer

Phases

Each phase is usable on its own; stop anywhere and it still works.

1. **City mode, no hardware:** backend on Proxmox, Tailscale, full-screen PWA with the Watch Dogs HUD, PTV, cameras, environment, City of Melbourne, buildings, planes and ISS via APIs, Broadcastify and KiwiSDR.
2. **Self-profile and personal layers:** profile card, gym mode at Hammers, uni, learner logbook, photography, home lab, grid and industry lens.
3. **First radio:** ADS-B on the roof, OpenWebRX+ on the phone, AIS.
4. **House mode:** Home Assistant VM, ESP32s with ESPresense, room cards, unknown-device alert, Shortcuts triggers.
5. **Dispatch transcription:** whisper.cpp, GPS-matched alerts, delivered through SPENCER.
6. **Sky and atmosphere:** Meteor-M images, radiosonde tracking and recovery.
7. **Amateur licence:** Foundation exam, APRS, WICEN.

Open questions and to-dos

- [ ] Apply for a PTV API key now; approval is not instant
- [ ] Order the SDR dongles, antennas and ESP32s so they arrive after exams
- [ ] Create the new GitHub repo, clone to E:, run /gsd-new-project on the PC
- [ ] LifeOS: what is it built on, and should Shinigami feed it, read from it, or replace part of it?
- [ ] Liftoff: can workouts be exported, or synced to Apple Health?
- [ ] Apple Watch: buy one for sleep and recovery data, or skip the recovery score?
- [ ] Check RadioReference: is CFA dispatch in your district analog, digital, or encrypted?
- [ ] Check VicRoads for the current learner logbook rules
- [ ] Register interest in the VicEmergency data feed
- [ ] Install GSD into the new repo (project-local) so cloud sessions get the commands too

Brief for /gsd-new-project

Copy this block whole into the first GSD question.

> **Shinigami Eyes**: a private, Watch Dogs-style profiler of my surroundings, for me alone. A full-screen iPhone home-screen web app (PWA) with a dark, glitchy HUD shows a live profile of where I am. Design rule: like Death Note's Shinigami eyes, everything shown gets a name and a countdown. **City mode (v1):** from GPS, nearby PTV stops with real-time departures, known CCTV within ~150 m (OpenStreetMap) and official road safety cameras, EPA air quality, UV, pollen, BOM weather, fire danger, City of Melbourne pedestrian counts, parking sensors, trees and development, heritage and Wikipedia for nearby buildings, VicEmergency incidents, planes overhead, ISS passes, ships in the bay, ACMA tower sites, and Victoria's live grid mix. **Personal layers:** a self-profile card; gym mode at Hammers Gym Nunawading (Liftoff, Apple Health via iOS Shortcuts); uni (Monash chem eng, Moodle deadlines, exam countdowns); learner-driver logbook; photography (golden hour, aurora alerts); home lab health; downtime. **House mode:** ESP32s with ESPresense and sensors through a Home Assistant VM: room cards, household presence, unknown-device alerts. **Radio:** RTL-SDRs on Proxmox (USB passthrough to a Debian VM) with OpenWebRX+, plus always-on readsb, AIS-catcher, radiosonde_auto_rx and SatDump; Broadcastify for CFA/FRV dispatch; local whisper.cpp transcription matched to my GPS for proximity alerts, delivered later by SPENCER. **Constraints:** backend on my Proxmox server, phone reaches it only over Tailscale; personal data never leaves Proxmox; receive-only, unencrypted signals only, no pager decoding, no police, nothing rebroadcast; nothing that identifies or tracks other people (Victoria's Surveillance Devices Act); iOS Shortcuts automations instead of background location; cached data on screen in under a second. Solo developer, Melbourne, summer holidays. No Home Assistant yet. Integrates with SPENCER 2.0 and my LifeOS dashboard later.

Sources

- Intercepting emergency service pager messages, Australian Emergency Law
- Using a scanner to listen to emergency radio, Australian Emergency Law
- Broadcastify Victoria feeds
- VicEmergency data feed access
- AusAlert rollout, Australian Security Magazine
- RTL-SDR weather satellite reception in 2026
- Tracking radiosondes with an RTL-SDR
- Get licensed, AHARS

- RTL-SDR V4 headless scanner on Proxmox, RadioReference
- Debian with RTL-SDR and OpenWebRX+

Other data sources in this brief (PTV, OpenStreetMap, City of Melbourne, AEMO, EPA, BOM) are from memory and still need checking at build time.